Karthik Reddy| AI/ML ENGINEER

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SUMMARY

AI/ML Engineer with 4 years of experience applying machine learning to real-world challenges in healthcare, finance, and telecom. Proven track record of designing and deploying systems that improved customer interaction efficiency by 30%+, reduced processing delays by 40%, and enhanced decision-making accuracy. Contributed to automation initiatives that eliminated 25%+ of manual tasks and strengthened compliance in high-volume data environments. Played a key role in streamlining document handling (↑ accuracy by 12%), fraud detection (↑ precision by 24%), and customer service workflows (↓ resolution time by 40%). Recognized for delivering production- grade, scalable AI solutions that align with organizational goals, boost cross-departmental productivity by 20%, and create measurable business impact.

TECHNICAL SKILLS

Languages & Scripting: Python, Java, C++, JavaScript, Bash
Machine Learning & AI: Regression, Classification, Clustering, SVM, XGBoost, Feature Engineering, Model Evaluation, Hyperparameter Tuning, scikit-learn
Deep Learning: Neural Networks, CNN, LSTM, TensorFlow, Keras, PyTorch, Reinforcement Learning
NLP & LLMs: OpenAI API, Hugging Face, LangChain, Prompt Engineering, NER, Text Summarization, Chatbot Development, RAG, Vector Databases
Data Engineering: SQL, NoSQL, ETL Pipelines, Data Cleaning, Data Modeling, Query Optimization, Apache Spark
Databases: MySQL, PostgreSQL, MongoDB, DynamoDB, Cassandra
Cloud & DevOps: AWS (EC2, S3, Lambda, SageMaker), Docker, Kubernetes, FastAPI, REST APIs, GitHub Actions, CI/CD, Terraform (basic)
Tools & Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scipy, Git, JSON, Regex
Simulation & Systems: Environment Simulation, Reward Engineering, LIDAR Object Detection
Security & Compliance: HIPAA, Secure Identity Protocols, Blockchain (Smart Chain)

PROFESSIONAL EXPERIENCE

Cohere **Feb 2025– Current**
AI/ML Engineer

- Designed and deployed end-to-end ML solutions using Python, scikit-learn, XGBoost, TensorFlow, PyTorch, and Keras, improving model accuracy by 25% and reducing prediction latency by 30%
- Developed NLP & LLM applications using OpenAI API, Hugging Face, and LangChain, including chatbots and RAG pipelines, increasing automation efficiency by 40% and response accuracy by 35%
- Built and optimized ETL pipelines, SQL/NoSQL queries, and data models across MySQL, PostgreSQL, MongoDB, DynamoDB, and Cassandra, reducing query latency by 30% and improving data quality by 25%
- Implemented reinforcement learning simulations with reward engineering and LIDAR-based object detection, improving decision accuracy by 28% and reducing processing time by 20%
- Automated deployments using Docker, GitHub Actions, FastAPI, and Terraform on AWS (EC2, S3, Lambda), reducing deployment time by 40% and improving system scalability by 35%
- Delivered secure AI workflows aligned with HIPAA and blockchain-based protocols, reducing security risks by 30% and ensuring 100% compliance across data pipelines
- Collaborated with cross-functional teams using Git, Pandas, NumPy, Matplotlib, and Regex, accelerating experimentation cycles by 35% and improving model iteration speed by 30%

CoRover.ai **April 2021– June 2023**
AI/ML Engineer

- Built and deployed multilingual conversational AI assistants (chat, voice, video) using NLP, LLMs, and RAG, improving user engagement by 40% across enterprise platforms
- Developed and fine-tuned domain-specific LLM pipelines (BharatGPT-style) using Hugging Face, LangChain, and vector databases, enhancing response accuracy by 35%
- Engineered scalable chatbot and voice bot solutions integrated with WhatsApp, IVR, and web channels, handling millions of user interactions with 99.9% uptime
- Designed and optimized ETL pipelines and data workflows using Python, SQL, and Apache Spark, reducing data processing latency by 30%
- Built and deployed REST APIs using FastAPI and containerized applications with Docker and Kubernetes, enabling scalable AI model serving on AWS
- Implemented MLOps workflows using MLflow, CI/CD pipelines, and monitoring tools, reducing model deployment time by 40% and improving reliability

EDUCATION

Master of Science in Computer and information science **May 2025**
UNIVERSITY OF NORTH TEXAS
Bachelors in Computer Science and Engineering, Artificial Intelligence **May 2023**
VELLORE INSTITUTE OF TECHNOLOGY

Project

Fraud Detection with RAG Pipelines | Hugging Face, LangChain, BERT, AWS SageMaker, FAISS, Pinecone

- Developed a Retrieval-Augmented Generation (RAG) system integrating BERT embeddings with FAISS and Pinecone to detect anomalous transaction patterns across 500K+ daily financial records.
- Reduced false positive fraud alerts by 24% by combining vector-based similarity search with domain-specific classification models.
- Implemented model training, versioning, and deployment workflows on AWS SageMaker, ensuring scalable and reproducible experimentation.
- Containerized and deployed the solution with Docker, enabling real-time inference and integration into existing fraud monitoring pipelines.
- Designed evaluation metrics to monitor precision, recall, and latency, improving model interpretability and operational reliability for risk teams.

IoT Predictive Maintenance & Financial Forecasting | *PyTorch, Apache Spark, Kafka, TensorRT, GCP Vertex AI, BigQuery, XGBoost, LightGBM, PostgreSQL, MongoDB, Tableau, Power BI*

- Built predictive maintenance models on IoT sensor data using PyTorch, Spark Streaming, and Kafka, deployed on GCP Vertex AI, reducing equipment downtime by 18%.
- Implemented financial time-series forecasting with ARIMA, Prophet, XGBoost, and LightGBM on PostgreSQL and MongoDB datasets, improving forecast precision for revenue and expenses.
- Optimized inference pipelines with TensorRT to deliver low-latency predictions for streaming IoT and financial data.
- Automated ETL workflows integrating structured and unstructured data into BigQuery, improving data consistency and query performance.
- Developed anomaly detection models to flag irregular sensor activity and financial outliers, supporting proactive maintenance and fraud risk monitoring.
- Delivered executive dashboards in Tableau and Power BI to visualize asset health, portfolio performance, and forecasting accuracy.

Computer Vision for Smart Surveillance | *OpenCV, YOLO, Detectron2, TensorFlow, Flask, Docker*

- Developed a real-time vision pipeline using YOLO and Detectron2 to detect intrusions, unattended objects, and abnormal activities, improving detection accuracy by 20%.
- Applied TensorFlow with transfer learning and augmentation to train models on diverse video datasets, enhancing performance in low-light and crowded environments.
- Integrated multi-camera stream processing with OpenCV, enabling simultaneous monitoring and object tracking across different feeds.
- Optimized inference workflows using GPU acceleration and Docker, reducing latency and ensuring scalable deployment.
- Exposed detection services through Flask APIs for seamless integration with enterprise security systems.
- Implemented automated alerting workflows that triggered notifications during anomalies, supporting faster incident response.